



POLITECNICO
MILANO 1863

SPACECRAFT ATTITUDE DYNAMICS AND CONTROL
Lab 4 – Attitude Kinematics

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First Tutoring

Thursday October 15th

Room BL27.1.5

Hour: 14.15 -16.15

Write questions to me within this afternoon:

annachiara.ippolito@polimi.it

These two hours are meant to fill the gaps on the fundamentals of **MATLAB** and **Simulink**.

Partecipate if you have questions, it will not be a lecture but an interactive session!

www.tutorapp.polimi.it -> try doing a first access!

Tomorrow you must register within the first half an hour.



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Task 1: Simulate kinematics using the DCM with standard integration

Principal moments of Inertia

$$I_x = 0.0700 \text{ kg m}^2$$

$$I_y = 0.0550 \text{ kg m}^2$$

$$I_z = 0.0250 \text{ kg m}^2$$

Initial conditions

$$\omega_x(0) = 0.45 \text{ rad/s}$$

$$\omega_y(0) = 0.52 \text{ rad/s}$$

$$\omega_z(0) = 0.55 \text{ rad/s}$$

$$A(0) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Governing Equations

Dynamics

$$\begin{cases} \dot{\omega}_x = \frac{I_y - I_z}{I_x} \omega_y \omega_z \\ \dot{\omega}_y = \frac{I_z - I_x}{I_y} \omega_x \omega_z \\ \dot{\omega}_z = \frac{I_x - I_y}{I_z} \omega_y \omega_x \end{cases}$$

Kinematics

$$\frac{dA(t)}{dt} = -[\omega]^\wedge A(t)$$

With

$$[\omega]^\wedge = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

Tasks

- Simulate the kinematics starting trying different values of the inertia tensor and initial conditions
- Test the orthonormality of the propagated solution

Task 2: Orthonormalize the matrix at each timestep

Traditional numerical integration procedures introduce numerical errors that affect the orthonormality of the solution. A possible solution is to use **structure-preserving integrators**. Another solution is to orthonormalize the solution at each timestep:

$$A_{k+1}(t) = \frac{3}{2}A_k(t) - \frac{1}{2}A_k(t)A_k^T(t)A_k(t)$$

This approach converges rapidly, with increasing k , to the exact value of A . In a first order approximation it is possible to adopt a single step iteration:

$$A(t) = \frac{3}{2}A_0(t) - \frac{1}{2}A_0(t)A_0^T(t)A_0(t)$$

Tasks

- Test the orthonormality of the propagated solution after orthonormalization

Task 3: Simulate kinematics using quaternions

Principal moments of Inertia

$$I_x = 0.0700 \text{ kg m}^2$$

$$I_y = 0.0550 \text{ kg m}^2$$

$$I_z = 0.0250 \text{ kg m}^2$$

Initial conditions

$$\omega_x(0) = 0.45 \text{ rad/s}$$

$$\omega_y(0) = 0.52 \text{ rad/s}$$

$$\omega_z(0) = 0.55 \text{ rad/s}$$

$$\mathbf{q}(0) = [0, 0, 0, 1]^T \text{ (Scalar last)}$$

Governing Equations

Dynamics

$$\begin{cases} \dot{\omega}_x = \frac{I_y - I_z}{I_x} \omega_y \omega_z \\ \dot{\omega}_y = \frac{I_z - I_x}{I_y} \omega_x \omega_z \\ \dot{\omega}_z = \frac{I_x - I_y}{I_z} \omega_y \omega_x \end{cases}$$

Kinematics

$$\dot{\mathbf{q}}(t) = \frac{1}{2} \Omega(t) \mathbf{q}(t)$$

with

$$\Omega(t) = \begin{bmatrix} 0 & \omega_z & -\omega_y & \omega_x \\ -\omega_z & 0 & \omega_x & \omega_y \\ \omega_y & -\omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix}$$

Tasks

- Simulate the kinematics starting trying different values of the inertia tensor and initial conditions
- Test the normality of the propagated solution